1 覚えるべき三角形の図示 解答

1. $30°$-$60°$-$90°$ の辺比 $1:\sqrt{3}:2$
2. $45°$-$45°$-$90°$ の辺比 $1:1:\sqrt{2}$
3. 最短辺 $=1 \Rightarrow (1,\sqrt{3},2)$
4. 斜辺 $=1 \Rightarrow \left(\frac{\sqrt{2}}{2},\frac{\sqrt{2}}{2},1\right)$
5. $\sin 30° = \frac{1}{2},\ \cos 30° = \frac{\sqrt{3}}{2},\ \sin 45° = \cos 45° = \frac{\sqrt{2}}{2},\ \sin 60° = \frac{\sqrt{3}}{2},\ \cos 60° = \frac{1}{2}$
6. $\tan 30° = \frac{1}{\sqrt{3}},\ \tan 45° = 1,\ \tan 60° = \sqrt{3}$

2 直角三角形の比 解答

1. $2,\ 2\sqrt{3},\ 4$
2. $4,\ 4\sqrt{3},\ 8$
3. $6,\ 6,\ 6\sqrt{2}$
4. $30°$側 $=3,$ 斜辺 $=6$
5. $60°$側 $=5\sqrt{3},$ 斜辺 $=10$
6. もう 1 辺 $=4,$ 斜辺 $=4\sqrt{2}$

3 三角比 解答

1. $\cos 30° = \frac{\sqrt{3}}{2},\ \cos 45° = \frac{\sqrt{2}}{2},\ \cos 60° = \frac{1}{2}$
2. $\sin 30° = \frac{1}{2},\ \sin 45° = \frac{\sqrt{2}}{2},\ \sin 60° = \frac{\sqrt{3}}{2}$
3. $\tan 30° = \frac{1}{\sqrt{3}},\ \tan 45° = 1,\ \tan 60° = \sqrt{3}$
4. 底辺 $=4\sqrt{3},$ 高さ $=4$
5. 底辺 $=5,$ 高さ $=5\sqrt{3}$
6. 斜辺 $=6\sqrt{2},$ 高さ $=6$

4 ラジアン 解答

1. $30° = \frac{\pi}{6},\ 45° = \frac{\pi}{4},\ 60° = \frac{\pi}{3},\ 90° = \frac{\pi}{2}$
2. $\frac{\pi}{6} = 30°,\ \frac{\pi}{4} = 45°,\ \frac{\pi}{3} = 60°,\ \frac{\pi}{2} = 90°$
3. $\cos\frac{\pi}{3} = \frac{1}{2},\ \sin\frac{\pi}{6} = \frac{1}{2},\ \tan\frac{\pi}{4} = 1$
4. 底辺 $=4\sqrt{3},$ 高さ $=4$
5. 底辺 $=2\sqrt{2},$ 高さ $=2\sqrt{2}$
6. 斜辺 $=6,$ 高さ $=3\sqrt{3}$

5 単位円の三角比 解答

1. $\sin\theta = \frac{\sqrt{3}}{2},\ \cos\theta = \frac{1}{2},\ \tan\theta = \sqrt{3}$
2. $\sin\theta = \frac{\sqrt{2}}{2},\ \cos\theta = \frac{\sqrt{2}}{2},\ \tan\theta = 1$
3. $\sin\theta = \frac{\sqrt{3}}{2},\ \cos\theta = -\frac{1}{2},\ \tan\theta = -\sqrt{3}$
4. $\cos\theta = \pm\frac{2\sqrt{2}}{3},\ \tan\theta = \pm\frac{\sqrt{2}}{4}$
5. $\sin\theta = \frac{\sqrt{15}}{4},\ \tan\theta = \sqrt{15}$
6. $\sin\theta = \frac{2\sqrt{5}}{5},\ \cos\theta = \frac{\sqrt{5}}{5}$

6 三角比の総合 解答

1. $\frac{1+\sqrt{3}}{2}$
2. $\frac{1}{2}$
3. $\frac{\sqrt{3}}{2}$
4. $\sqrt{3}+2$
5. 1
6. $\cos\theta = \frac{4}{5},\ \tan\theta = \frac{3}{4}$